

Exercise 7 : Financial Forecasting

Recursion

Recursion is a technique where method calls itself to solve a problem by breaking it into smaller subproblems.

Recursive Formula Used

$\text{Forecast}(0) = \text{Initial Value}$

$\text{Forecast}(n) = \text{Forecast}(n - 1) * (1 + \text{Growth rate})$

This implementation satisfies the exercise requirements by:

- Using recursion
- Including a base case
- Applying the recursive formula
- Demonstrating financial forecasting with sample values

Exercise7Main.java

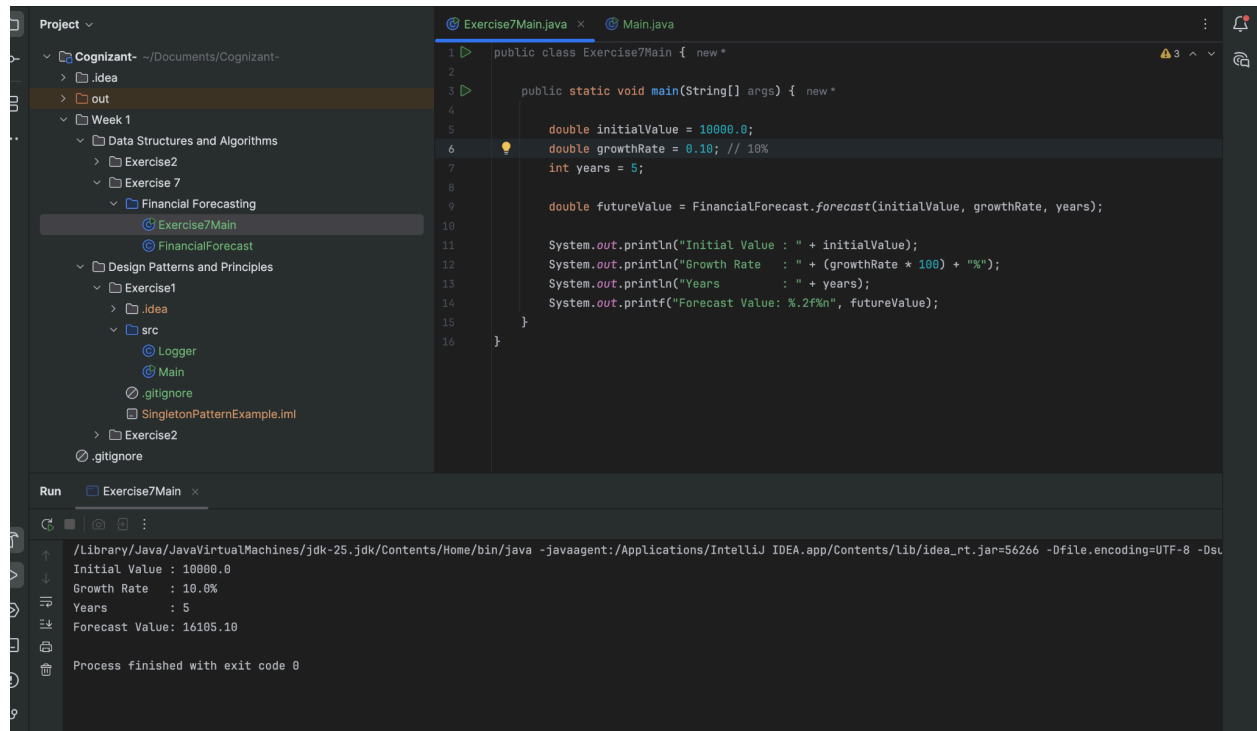
```
public class Main {  
    public static void main(String[] args) {  
        double initialValue = 10000.0;  
        double growthRate = 0.10; // 10%  
        int years = 5;  
        double futureValue = FinancialForecast.forecast(initialValue, growthRate, years);  
        System.out.println("Initial Value : " + initialValue);  
        System.out.println("Growth Rate  : " + (growthRate * 100) + "%");  
        System.out.println("Years      : " + years);  
        System.out.printf("Forecast Value: %.2f%n", futureValue);  
    }  
}
```

```
}  
}
```

FinancialForecast.java

```
public class FinancialForecast {  
  
    // Recursive method to calculate future value  
    public static double forecast(double initialValue, double growthRate, int years) {  
  
        // Base case  
        if (years == 0) {  
            return initialValue;  
        }  
  
        // Recursive case  
        return forecast(initialValue, growthRate, years - 1) * (1 + growthRate);  
    }  
}
```

Output:



Time Complexity:

- Recursive Solution: $O(n)$
- Space Complexity: $O(n)$ (due to the recursion stack)